BioMedIT: Switzerland's secure IT network for responsible health data processing

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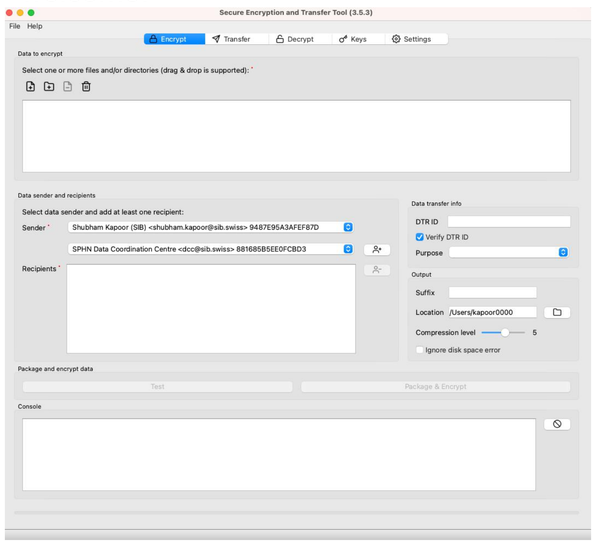
Connecting researchers from across Switzerland with biomedical data to foster personalized health: This is the aim of the national secure computing network BioMedIT, set up by the SIB Swiss Institute of Bioinformatics in collaboration with ETH Zurich, University of Basel, and the University of Lausanne. Launched in 2017 as part of the Swiss Personalized Health Network (SPHN) initiative, the network is now fully operational with over 60 national and international health-related research projects running, from infection prediction to precision oncology.

>BIOMED IT< sett

BioMedIT secure encryption and transfer tool

- Standardized data package and transfer from data providers to BioMedIT nodes
- Automation around GPG, tar, SFTP, etc.
- CLI and GUI version
- OpenSource & reusable, contributions welcome!

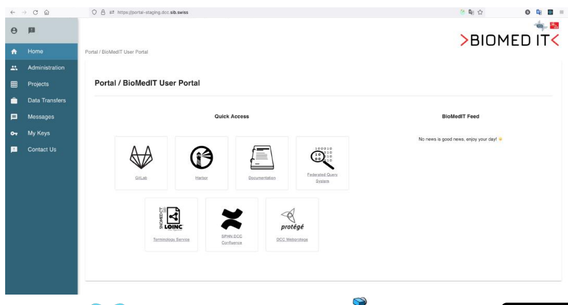


GnuPG, SFTP, LiquidFiles, Portal git

>BIOMED IT< Portal

BioMedIT Portal: The one stop shop for accessing BioMedIT`s secure services

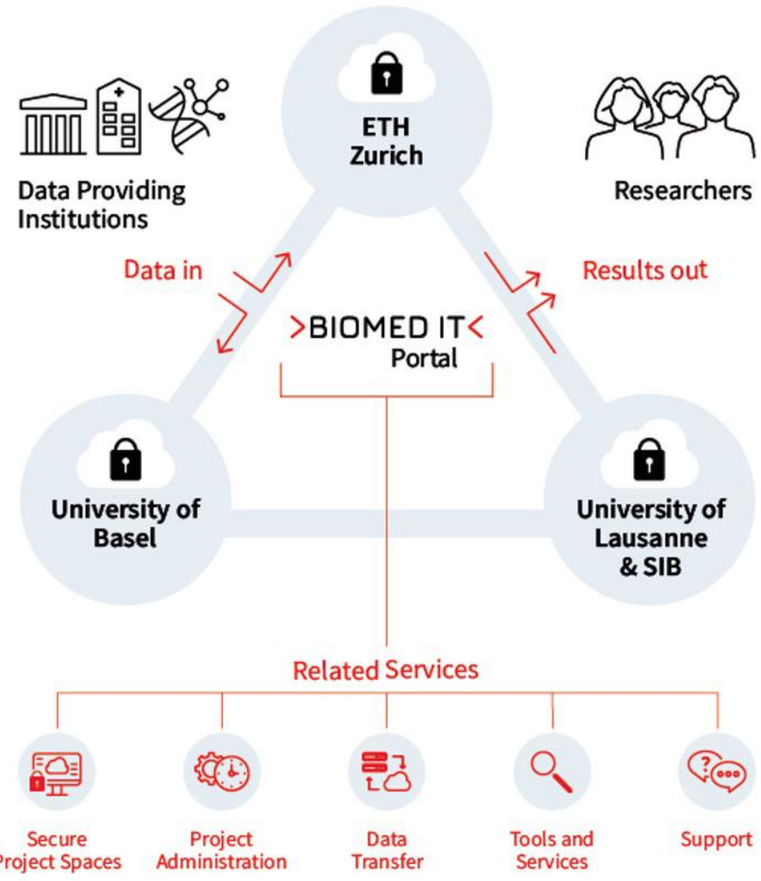
- Project administration
- User management
- Data Transfer Request management
- Access link to B-Spaces
- OpenSource & reusable, contributions welcome!



React, Docker, django, Portal git

BioMedIT in numbers

- Aggregated Compute: 5K+ Cores, 425+ GPUs, 2.5+ PB storage
- Registered Users: 344
- Onboarded Data Providers: 24
- Production Data Transfers: 58
- Total Projects: 60 (national and intern.)



BioMedIT Nodes

sciCORE

Universität Basel

OpenStack
1.0 PB Storage
Compute Capacity:
•500 CPUs
•8 GPUs
•5 TB RAM
HPC Scheduler: SLURM
File System: CEPH, GlusterFS

LabKey, openstack, slurm

SENSA

UNIL | Université de Lausanne, SIB Swiss Institute of Bioinformatics

OpenStack
400TB Storage
Compute Capacity:
•800 CPUs
•6 GPUs
•3 TB RAM
HPC Scheduler: SLURM
File System: Weka, CEPH, Scality

WEKA, openstack, slurm

SIS

ETH zürich

OpenStack
2.0 PB Storage
Compute Capacity:
•4400 CPUs
•424 GPUs
•41 TB RAM
HPC Scheduler: SLURM
File System: Luster file storage

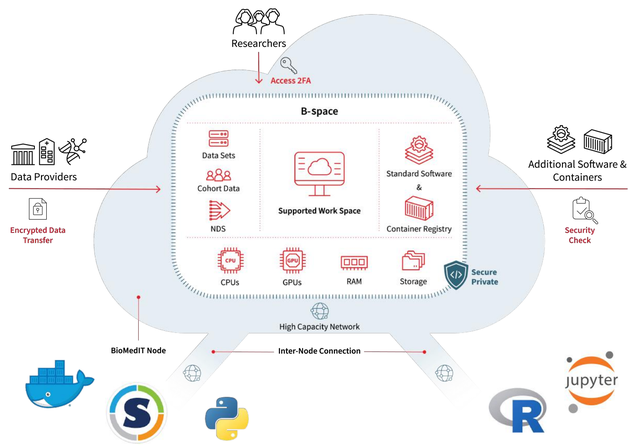
openBIS, openstack, slurm

A unified approach to security

>BIOMED IT< B-spaces

BioMedIT secure research project spaces

- Multi-tenancy setup: isolated project storage and compute, remote desktops
- Services accessible via *Strong* Authentication



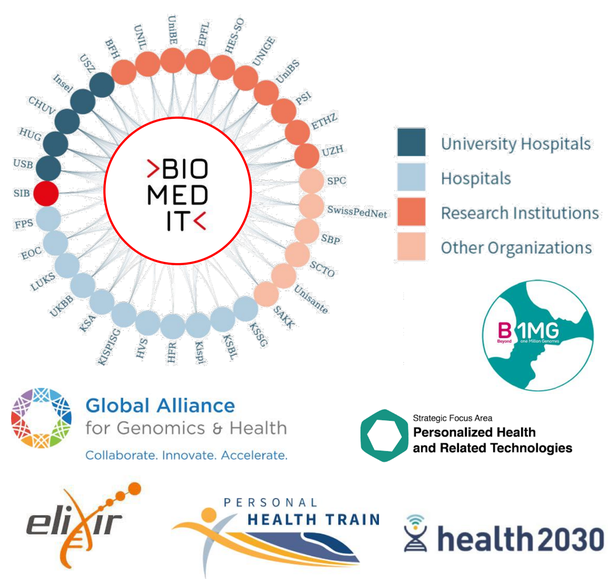
>BIOMED IT< Central Services

- DCC Git: Secure code management service with CI/CD possibility
- DCC Container Registry: Secure registry for OCI container management
- DCC WebProtégé: Web-based, collaborative ontology development environment
- DCC Terminology Service: Provision of SPHN compatible versions of Swiss & international terminologies
- DCC Keyserver: GPG key management
- DCC Confluence: Secure collaborative space for users and working groups

protégé, GitLab, SNOMED CT, LOINC, Tools & Resources

Collaborations

BioMedIT enables national and intern. research collaborations



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